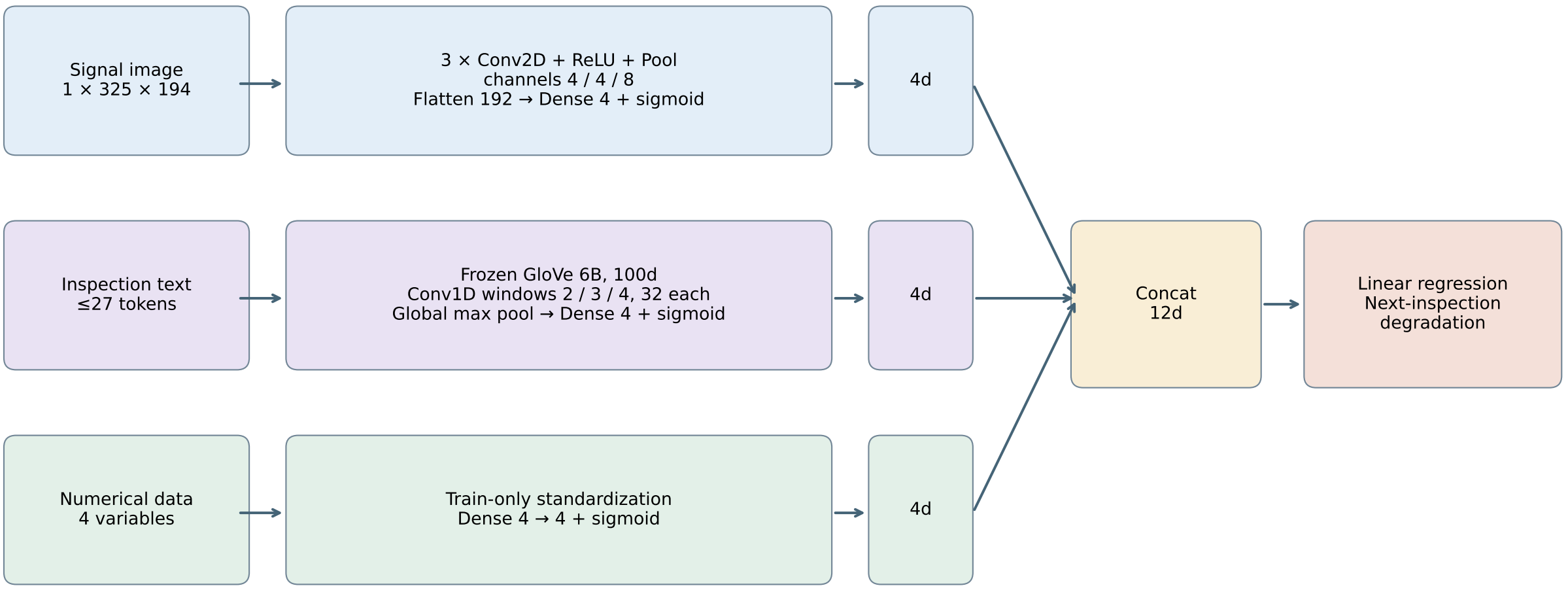
MB-DNN: multi-branch multimodal degradation prediction
Signal image
1 × 325 × 194
3 × Conv2D + ReLU + Pool
channels 4 / 4 / 8
Flatten 192 → Dense 4 + sigmoid
4d
Inspection text
≤27 tokens
Frozen GloVe 6B, 100d
Conv1D windows 2 / 3 / 4, 32 each
Global max pool → Dense 4 + sigmoid
4d
Numerical data
4 variables
Train-only standardization
Dense 4 → 4 + sigmoid
4d
Concat
12d
Linear regression
Next-inspection
degradation
Train-only scaling and vocabulary • MSE training • Validation-based checkpoint selection